

ALAMEIN INTERNATIONAL UNIVERSITY

Faculty of Computer Science & Engineering

Knowledge Based Systems | AIE212

Medical Diagnosis Expert System

Clinical Triage & Preliminary Diagnosis KBS

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PHASE 3: DESIGN PHASE

3.1 Knowledge Base Design

The Knowledge Base (KB) is the core repository of all domain expertise in the system. It stores medical knowledge in four interconnected layers: symptoms, diseases/conditions, patient history, and risk factors. Each layer serves a distinct role in the reasoning process, and the layers interact through structured relationships that the inference engine traverses during diagnosis.

Types of Knowledge Stored

Knowledge Type	What It Contains	Role in Reasoning
Symptoms	Presenting complaints (chest pain, fever, dyspnea, nausea, etc.), sub-types (crushing, sharp, burning), location, radiation, associated signs	Matched against rule antecedents as primary input facts
Diseases / Conditions	~40 conditions across cardiovascular, respiratory, neurological, GI, urological, and general medicine categories; each with ICD-10 reference and urgency class	Form the hypothesis space — rule consequents derive these as candidate diagnoses
Patient History	Age, sex, BMI, known chronic conditions (diabetes, hypertension, asthma, cardiac disease), current medications, allergies, smoking/alcohol status	Contextual modifiers that raise or lower certainty factors for relevant rules
Risk Factors	Hypertension, diabetes mellitus, obesity, smoking, sedentary lifestyle, family history of cardiac disease	Pattern-amplifiers: when co-occurring with symptoms, they boost CF scores on associated rules
Vital Signs	SpO2 (%), heart rate (bpm), blood pressure (mmHg), body temperature (°C), respiratory rate (breaths/min), level of consciousness	Quantitative thresholds that trigger red-flag rules or override urgency classification
Clinical Actions	Emergency calls, medication suggestions (aspirin 300 mg), referral pathways, nil-by-mouth orders, GP follow-up instructions	Fired by action rules after a diagnosis and urgency level have been confirmed

Relationships Between Entities

Knowledge entities are linked through directional relationships that guide inference from patient inputs to clinical recommendations:

- **Symptom → Condition:** One or more symptoms support a condition with an associated certainty factor. For example, chest pain + diaphoresis + left-arm radiation supports Acute MI with CF = 0.95.
- **Condition → Action:** A confirmed diagnosis maps to a set of recommended clinical actions. Acute MI maps to [call emergency services, aspirin 300 mg if not contraindicated].
- **Risk Factor → CF Modifier:** Patient background factors (e.g., hypertension, diabetes) amplify the CF of cardiovascular and neurological rules. Hypertension boosts cardiac rules by +0.10.
- **Vital Sign → Urgency Override:** Certain vital thresholds (SpO2 < 90%, unresponsive patient) directly override urgency to IMMEDIATE regardless of the active diagnosis.
- **Red Flag → Emergency Shortcut:** When a defined red-flag pattern is detected (e.g., crushing chest pain + diaphoresis), the inference process short-circuits to emergency protocol immediately.

Example Knowledge Base Entries

Below are three example entries illustrating how different knowledge types are encoded in the KB:

KB Entry 001 — Condition Node: Acute Myocardial Infarction

```
CONDITION : Acute_Myocardial_Infarction
category  : Cardiovascular | Red Flag
urgency    : IMMEDIATE
icd_ref    : I21.9
base_CF    : 0.95
linked_rules : [R-08, R-09]
actions    : [call_emergency_services, aspirin_300mg_if_not_contraindicated]
differentials: Unstable_Angina (CF 0.72), Pulmonary_Embolism (CF 0.55)
```

KB Entry 002 — Symptom Node: Chest Pain

```
SYMPTOM    : chest_pain
sub_types  : [crushing, sharp, burning, pressure, tearing]
radiation  : [left_arm, jaw, back, epigastric, shoulder]
linked_to  : [Acute_MI, Unstable_Angina, Pulm_Embolism, Aortic_Dissection, GERD]
red_flag_if: sub_type IN [crushing, pressure] AND radiation IN [left_arm, jaw]
```

KB Entry 003 — Risk Factor: Hypertension

```
RISK_FACTOR : hypertension
CF_boost    : +0.10 on cardiovascular rules (Acute_MI, Heart_Failure)
CF_boost    : +0.08 on stroke rules (Ischemic_Stroke, TIA)
linked_conditions : [Acute_MI, Stroke, Hypertensive_Crisis, Heart_Failure]
```

3.2 Knowledge Representation

All medical knowledge in the system is encoded using IF–THEN production rules. Each rule has a condition part (antecedent) listing the facts that must be true, and a conclusion part (consequent)

containing the derived fact with its certainty factor, urgency level, and recommended actions. This mirrors the clinical reasoning patterns identified in Phase 2 Section 2.1.

Simple Rules — Single Symptom → Condition

Simple rules fire when a single defining symptom or vital sign is present. They carry lower CF values because a single finding is rarely diagnostic on its own, but they seed the Working Memory for more complex rules to act on.

Rule R-01 — Elevated Temperature → Infection Present

```
IF body_temperature > 38.0°C
THEN hypothesis = Infection_Present [CF = 0.55]
AND urgency_candidate = SEMI_URGENT
```

Rule R-02 — Critical SpO2 → Respiratory Emergency

```
IF spo2 < 90%
THEN red_flag = SpO2_Critical [CF = 1.0]
AND urgency = IMMEDIATE
AND action = [administer_O2, call_emergency_services]
```

Rule R-03 — Possible Influenza (Fever + Cough)

```
IF symptom = fever AND symptom = cough
THEN hypothesis = Possible_Influenza [CF = 0.50]
```

Complex Rules — Multiple Symptoms → Condition

Complex rules combine several facts — symptoms, vitals, history, and risk factors — to reach high-confidence diagnoses. These are the primary drivers of differential diagnosis generation, as established in Phase 2 Section 2.1.

Rule R-08 — Red Flag: Suspected Acute Myocardial Infarction

```
IF patient_age >= 40
AND symptom = chest_pain
AND pain_character IN [crushing, pressure]
AND pain_radiation IN [left_arm, jaw]
AND symptom = diaphoresis
AND (spo2 < 95 OR heart_rate > 100)
THEN diagnosis = Acute_Myocardial_Infarction [CF = 0.95]
AND urgency = IMMEDIATE
AND red_flag = TRUE
AND action = [call_emergency_services, aspirin_300mg]
```

Rule R-14 — Red Flag: Suspected Ischemic Stroke (FAST Protocol)

```
IF symptom = facial_drooping
AND symptom = arm_weakness
AND symptom = slurred_speech
AND onset_type = sudden
AND symptom_duration <= 4.5 hours
THEN diagnosis = Ischemic_Stroke [CF = 0.97]
AND urgency = IMMEDIATE
AND action = [call_emergency_services, nil_by_mouth, thrombolysis_window_active]
```

Rule R-05 — Probable Influenza (Multi-Symptom)

```

IF    symptom = fever    (temperature > 38°C)
AND   symptom = cough
AND   symptom = myalgia
AND   onset_type = sudden
THEN diagnosis = Influenza [CF = 0.74]
AND   urgency = NON_URGENT
AND   action = [rest, fluids, OTC_analgesics, GP_if_worsens]

```

Rule R-22 — Probable Lower Urinary Tract Infection

```

IF    patient_sex = female
AND   symptom = dysuria
AND   symptom = urinary_frequency
AND   symptom_duration >= 1 day
AND   body_temperature < 38.5°C
AND   symptom != flank_pain
THEN diagnosis = UTI_Lower [CF = 0.82]
AND   urgency = SEMI_URGENT
AND   action = [urine_dipstick, empirical_antibiotics, GP_within_24h]

```

Risk-Factor Modifier Rules

These rules do not derive diagnoses directly. Instead, they boost the CF of hypotheses already in Working Memory when relevant patient history is detected:

Rule RF-01 — Hypertension Amplifier

```

IF    known_condition = hypertension
AND   hypothesis IN [Acute_MI, Stroke, Hypertensive_Crisis]
THEN CF(hypothesis) += 0.10

```

Rule RF-02 — Diabetes Amplifier

```

IF    known_condition = diabetes_mellitus
AND   hypothesis IN [Acute_MI, Sepsis, Diabetic_Ketoacidosis]
THEN CF(hypothesis) += 0.08

```

3.3 Handling Multiple Possible Conditions

In clinical medicine, a patient's presentation rarely points unambiguously to a single condition. The system handles this reality through a differential diagnosis framework built on Certainty Factors, returning a ranked list of candidates rather than a single answer.

Returning Multiple Diagnoses

When the inference engine completes its cycle, all diagnostic hypotheses derived from fired rules are collected into a Differential List. Each entry includes the condition name, its aggregated CF score, the urgency level assigned by the highest-priority rule that contributed to it, and an evidence list showing which rules and facts were involved. This list is then passed to the Differential Consolidation module.

Assigning Confidence Levels

Certainty Factors are combined using the MYCIN calculus when multiple rules support the same diagnosis:

CF Combination Formula (two independent rules supporting the same diagnosis):

$$CF_{combined} = CF1 + CF2 \times (1 - CF1)$$

Example: Rule R-08 fires (CF = 0.95), Risk Factor RF-01 boosts (+0.10):

$$CF_{combined} = 0.95 + 0.10 \times (1 - 0.95) = 0.955$$

Confidence labels displayed to the clinician:

CF $\geq$ 0.85 → Very High

CF $\geq$ 0.70 → High

CF $\geq$ 0.50 → Medium

CF < 0.50 → Low

Ranking Conditions by Likelihood and Severity

The Differential Consolidation module ranks the hypothesis list using a two-pass strategy that prioritises patient safety over raw probability:

- **Pass 1 — Safety-First Sort:** Any diagnosis carrying IMMEDIATE urgency is always ranked first, regardless of its CF score. A lower-probability life-threatening condition outranks a high-probability minor one.
- **Pass 2 — CF-Based Sort:** Among diagnoses at the same urgency level, they are ordered by descending CF score. The final output shows the primary diagnosis plus up to three differentials.

Example differential output for a 58-year-old male with crushing chest pain, diaphoresis, and known hypertension:

Rank	Diagnosis	CF Score	Confidence	Urgency
1 — Primary	Acute Myocardial Infarction	0.955	Very High	IMMEDIATE
2	Unstable Angina	0.680	High	URGENT
3	Pulmonary Embolism	0.420	Medium	URGENT
4	Severe GERD with Referred Pain	0.210	Low	SEMI-URGENT

3.4 Handling Conflicting Conclusions

Conflicts arise when symptoms overlap across diseases (e.g., fever + chest pain could indicate pneumonia, pericarditis, or pulmonary embolism simultaneously) or when rules derive contradictory urgency levels. The system resolves these through a layered conflict resolution strategy, consistent with the agenda-based approach described in Phase 2 Sections 2.4 Steps 4–5.

Strategy	When Applied	How It Resolves the Conflict
Rule Priority (Specificity)	Multiple rules match at the same time	More specific rules win: a rule with 6 conditions outranks a rule with 2 conditions targeting the same diagnosis. This is the primary conflict criterion.
Certainty Factors	Same diagnosis supported by multiple rules	CFs are combined using MYCIN calculus. The aggregated score determines ranking in the differential list.
Urgency Promotion	Rules assign different urgency levels to the same diagnosis	Highest urgency always wins. A later non-urgent rule cannot downgrade an IMMEDIATE flag once it has been set.
Red Flag Override	A red-flag pattern is detected during inference	Inference halts immediately. Output switches to the emergency protocol regardless of any other active rules or lower-priority diagnoses.
Weighted Scoring	Overlapping symptom sets across multiple candidate diagnoses	Each candidate accumulates a score based on how many of its required conditions are met. Non-matching required conditions reduce the score proportionally.
Recency Preference	Two rules have identical priority, specificity, and CF score	The rule matching the most recently entered patient facts is preferred. This reflects the clinical intuition that the latest information is most relevant.

Worked Conflict Example

A 45-year-old male presents with fever (38.8°C), chest pain, and shortness of breath. Three rules fire simultaneously:

Conflict Scenario: Three Rules Fire Simultaneously

Rule R-11 → Pneumonia [CF = 0.71, urgency = URGENT, 3 conditions met]

Rule R-18 → Pulmonary Embolism [CF = 0.65, urgency = URGENT, 3 conditions met]

Rule R-08 → Acute MI [CF = 0.82, urgency = IMMEDIATE, 4 conditions met]

Resolution:

Pass 1 (Urgency Promotion) : Acute MI elevated to Rank 1 – IMMEDIATE urgency

Pass 2 (Specificity) : Acute MI also wins on conditions matched (4 vs 3)

Pass 3 (CF Sort at URGENT) : Pneumonia (0.71) outranks Pulm. Embolism (0.65)

Final: Primary = Acute MI (IMMEDIATE), Differentials = Pneumonia, PE (URGENT)

3.5 System Architecture Design

The system follows a classic Knowledge-Based System architecture with five major components. All components communicate through the Working Memory, which acts as the shared session state for the current patient assessment. The architecture directly implements the nine-step decision flow described in Phase 2 Section 2.4.

Component	Role	Phase 2 Steps Handled
User Interface (UI)	Collects and validates patient data; displays final diagnosis, differentials, urgency badge, red flag alerts, and reasoning explanation	Steps 1, 9
Knowledge Base	Stores all IF–THEN rules (~150+), fact reference tables, CF metadata, and action mappings for ~40 conditions	Referenced in Steps 3–6
Working Memory	Session-scoped store of all current patient facts (asserted inputs + derived conclusions); cleared after each assessment	Steps 2, 6
Inference Engine	Pattern-matches facts against rules, manages the agenda, resolves conflicts, fires rules, and writes conclusions back to WM using forward and backward chaining	Steps 3–7
Explanation Module	Records rule traces and generates plain-language summaries of reasoning for clinician review	Steps 7–9

User Interface (UI)

The UI is the entry point for all patient data. It presents a structured clinical intake form covering chief complaint, vital signs, patient background, and symptom detail — matching the input categories defined in Phase 1 Section 1.2. Before submission, all inputs are validated: numeric values are range-checked (e.g., SpO2 must be 50–100%), required fields are flagged, and contradictory entries are highlighted.

The output view renders the primary diagnosis with its urgency badge (color-coded: red for IMMEDIATE, amber for URGENT, yellow for SEMI-URGENT, green for NON-URGENT), the differential list with CF scores and confidence labels, red flag alerts, recommended actions, and the reasoning trace. The UI also supports three care-setting modes — GP Clinic, Telemedicine, and Emergency Department — which adjust which referral pathways are shown.

Knowledge Base

The Knowledge Base is organized into three sub-stores: the Rule Store containing all IF–THEN production rules with CF values, urgency classes, priority weights, and action pointers; the Fact Store holding static reference data such as vital-sign normal ranges, drug contraindication tables, and age-specific risk thresholds; and the CF Metadata Store containing CF boost tables indexed by risk factor and condition pair, along with the MYCIN combination formulas. Rules are written in human-readable form so

a domain expert can update them without modifying the inference engine — a key advantage of RBES identified in Phase 2 Section 2.2.

Working Memory (WM)

Working Memory is a session-scoped volatile store. At the start of each assessment, validated UI inputs are converted into structured fact triples (attribute, value, certainty) and loaded into WM. As rules fire, their conclusions are written back as new facts, which may trigger further rules in subsequent inference cycles. A dedicated Red Flag Register within WM tracks any red flags that have been raised; once set, a red flag cannot be cleared by later rules. WM is fully cleared at the end of each assessment, consistent with the snapshot model described in Phase 1 Section 1.5.

Inference Engine

The Inference Engine is the computational core of the system. It operates in two chaining modes, matching the hybrid reasoning approach described in Phase 2 Section 2.3:

- **Forward Chaining (Primary Mode):** Data-driven. Starts from asserted patient facts in WM, matches rule antecedents using a RETE network pattern matcher, adds matching rules to the Agenda, fires the top-priority rule, writes its conclusion to WM, and repeats until no new facts can be derived.
- **Backward Chaining (Verification Mode):** Goal-driven. Activated when a clinician wants to confirm a specific hypothesis, or when the system proactively checks for dangerous conditions after detecting relevant symptoms (e.g., triggering a stroke check when slurred speech is reported).

The Agenda Manager maintains a prioritized queue of rules ready to fire, ordered by: (1) red-flag rules first, (2) higher CF, (3) more specific rules (more conditions), (4) higher urgency class. When multiple rules support the same hypothesis, the CF Combiner applies the MYCIN formula before writing the merged fact to WM.

Explanation Module

The Explanation Module provides the transparent reasoning traces that are essential for clinical trust and regulatory compliance — a requirement identified in Phase 1 Section 1.3. It records every rule that fired during the session in firing order, with the specific patient facts that matched each antecedent. This trace is then converted into a plain-language narrative for the clinician, differential justifications (explaining why each alternative was included or ranked below the primary), red-flag annotations (identifying exactly which finding triggered each alert), and a log of any backward-chaining queries made during verification.

3.6 How the System Will Explain Its Decisions:

The system will include a dedicated **Explanation Module** designed to provide clear, transparent, and clinically meaningful reasoning for every decision it makes. This is essential because medical systems must avoid “black-box” behavior and allow healthcare professionals to understand and verify the logic behind each output.

The explanation mechanism will rely on **rule tracing**, **certainty factor tracking**, and **fact justification** generated during the inference process.

First, during execution, the inference engine will **record all fired rules** along with their conditions and outcomes. Each time a rule is triggered, the system logs:

- The rule ID (e.g., R-08 for suspected Acute MI)
- The input facts that satisfied the rule conditions
- The resulting conclusion (diagnosis, urgency, or action)
- The associated certainty factor (CF)

This creates a structured reasoning path that can later be translated into human-readable explanations.

Second, the system will explain decisions in a **step-by-step narrative format**. Instead of showing raw rules, the Explanation Module converts them into natural language. For example:

- “The patient reported chest pain radiating to the left arm and sweating.”
- “Vital signs show elevated heart rate and reduced oxygen saturation.”
- “These findings match the criteria for suspected Acute Myocardial Infarction.”
- “Due to the severity of these symptoms, the case is classified as Immediate.”

Third, the system will justify **multiple possible diagnoses** by showing how each one was supported. For every diagnosis in the ranked differential list, the explanation will include:

- Key supporting symptoms and signs
- The rules that contributed to that diagnosis
- The final combined certainty factor (e.g., High / Medium / Low)

This ensures that clinicians understand not only the top diagnosis but also alternative possibilities.

Fourth, in cases of **conflicting conclusions**, the system will explicitly explain how conflicts were resolved. It will state:

- Which rules competed
- Why one rule was prioritized (e.g., higher CF, more specific conditions, or higher urgency)
- That patient safety rules (red flags) override all others

For example:

- “Although symptoms suggest both UTI and appendicitis, appendicitis was prioritized due to higher severity and risk.”

Fifth, the system will provide **red flag explanations** when triggered. These will clearly highlight critical findings:

- “Red Flag Triggered: Oxygen saturation below 90%”
- “This indicates possible respiratory failure and requires immediate action.”

Sixth, the Explanation Module will support **“Why” and “How” queries**:

- **Why this diagnosis?** → Shows rules and evidence that led to it
- **Why not another diagnosis?** → Explains missing or weak conditions
- **How was this conclusion reached?** → Displays the reasoning chain step-by-step

Finally, the explanation output will be presented in **two levels of detail**:

- **Summary Explanation (default):** Simple, readable explanation for quick understanding

- **Detailed Explanation (optional):** Full rule trace, CF values, and reasoning steps for advanced users or debugging

This design ensures that the system remains:

- **Transparent** (users can see how decisions are made)
- **Trustworthy** (clinicians can verify logic)
- **Educational** (helps users understand clinical reasoning)